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Brac University

Semester: Fall 2025

Course Code: CSE430

Digital Signal Processing

Faculty: EHQ

Assessment: *Quiz 5*

Duration: 25 minutes

Date: January 6, 2026

Full Marks : 20

Decibel is a comparative unit of power where the conversion is given by the formula,

$$dB = 10 \cdot \log_{10} \frac{P}{P_{ref}} = 10 \cdot \log_{10} \frac{V^2}{V_{ref}^2} = 20 \cdot \log_{10} \frac{V}{V_{ref}}$$

■ Question 1 of 3 [6 marks]

After an FFT operation, the spectrum magnitude observed was,

$$||X[k]|| = \{\underline{4}, 7, 6, 9, 11, 8, 4, 2, 2, 1, 2, 2, 4, 8, 11, 9, 6, 7\}$$

If $f_s = 18 \cdot (\text{mod}(\text{ID}, 3) + 1)$, answer the following questions -

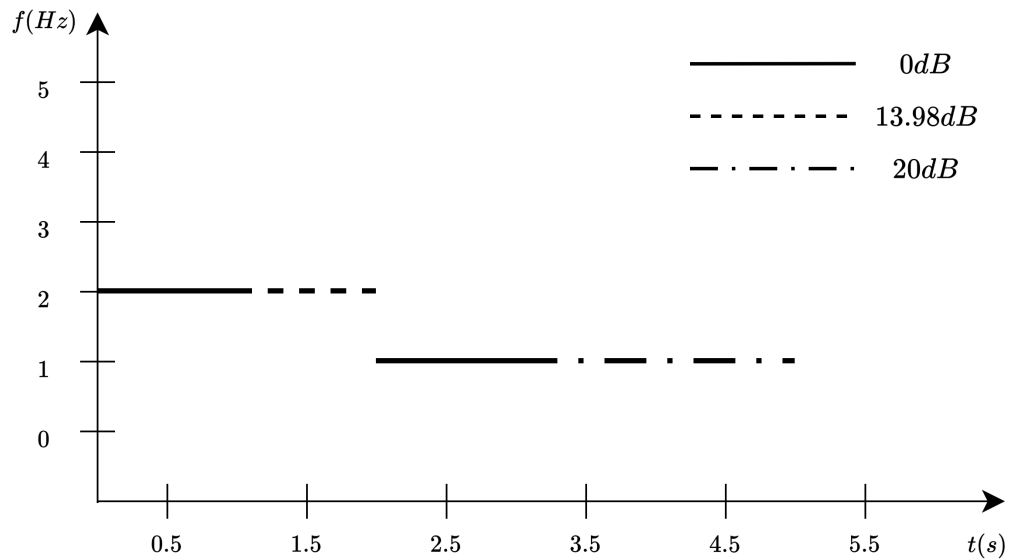
- [3 marks] Find the true frequency labels. *It is enough to write the frequency values as an array. You do not have to draw the signal in a plot.*
- [3 marks] Prepare the spectrum for its integration into a spectrogram, $||X'(f)||$. *You do not have to draw the spectrum. But in that case, include the corresponding frequencies in the updated spectrum as an array.*

■ Question 2 of 3 [6 marks]

Sharadia is working with a 5 minute audio signal sampled at $44kHz$. To extract high-level features from the signal, she has derived the spectrogram of the audio file. She used a Hann window of length equal to $5 \cdot k$ s with a window overlap of 40%. If $k = \text{mod}(\text{ID}, 3) + 1$, answer the following questions -

- (a) [2 marks] What will be the maximum frequency information available per segment? Provide reasoning.
- (b) [4 marks] Find the frequency and time dimension of the spectrogram. Add reasoning and show calculations where applicable.

■ Question 3 of 3 [8 marks]



The spectrogram has been prepared using a centered window of window length 1 second with no overlap. Assume no window contamination. The $\|X(f, t)\|$ is provided in decibel (dB). For this spectrogram, $V_{ref} = 2 \cdot k \text{ V}$ where $k = \text{mod}(\text{ID}, 3) + 1$. With these info, draw the approximate envelope of $x(t)$ with amplitude levels in voltage.

The amplitude scale doesn't have to be super accurate as variations maybe quite high.